

02

Insights Engine

From field insights to care and knowledge gaps to curated HCP content for each persona and stage.

Hackathon vertical 02 of 04 · build afternoon agenda, swarm, gates, question bank and impact worksheet

Facilitators: John and Anjali (insights pillar leads)

AI in Action for Medical Affairs

October 13–14, 2026 · Convene Philadelphia (2nd floor), Two Commerce Square, 2001 Market St

Host and keynote: Vivek Mukhatyar · **aiinaction.up.railway.app/hackathon**

Scan for the hackathon pack

02 AT A GLANCE

The vertical on one page

THE CHALLENGE

Field insights, enquiries and needs assessments point to the same care and knowledge gaps, but the signal rarely reaches the content plan. A community specialist new to the data and an academic expert get the same deck.

DAY 2 MORNING · WED OCT 14

2nd of 4 to present

01 **02** 03 04

About 20 min, including a few questions. Order 01→04, after the final build.

THE PROBLEM PROMPT

> Build an AI workforce that reads field observations, enquiries and the needs assessment, raises safety findings first, and names care and knowledge gaps (source IDs, confidence). It segments HCPs by persona and knowledge stage (aware, exploring, applying, expert), maps gaps to those cells, and curates a draft content plan: reuse, adapt or create, with the gap each item closes. Humans approve gaps and segments; MLR reviews new items.

Teams can keep this prompt or narrow it to one workflow they know. A good narrowed prompt names the role, the trigger, the deliverable and the decision it serves.

DONE LOOKS LIKE · DAY 1

The swarm ran end to end at least once on practice data; it stopped at Gate 1 and Gate 2 and the team made both decisions; screenshots and minute-level timings are in the capture log; the agent has drafted a slide outline.

DONE LOOKS LIKE · DAY 2

The deck is built from the final template: no [brackets] left, impact numbers are the team's own, practice data marked fictional, the LIVE DEMO slide left empty, backup screenshots in place. The team presents second of four, for about 20 minutes with a short live demo.

LEAD SKILL

`field-insight-synthesis`

SUPPORTING SKILLS

`insight-generation`

`evidence-gap-analysis`

`spreadsheet-analysis`

DATA PACK

Any TA; synthetic data (fictional Nordvant Biopharma) with seeded safety findings and a promotional objective to catch.

`field-observations.csv` `medical-plan.md`

`medical-education-needs.md`

`medical-information-enquiries.csv` + practice CRM

Fictional practice data (Nordvant Biopharma), or bring your own non-confidential data.

WARM-UP MISSIONS

L2 `field-insights`
What should leadership know?

L1 `medical-information`
Handle the enquiry queue

02 DAY 1 AFTERNOON · TUE OCT 13

The build afternoon, block by block

A suggested, optional order of four blocks with approximate durations; teams may move faster or slower. **No formal share-out at the end of the afternoon:** facilitators circulate and ask questions (page 6).

1 · Setup **about 30 min** → 2 · Warm-ups **about 40 min** → 3 · Ideation **about 45 min** → 4 · The hack **about 80 min** → **Then** cocktail reception

1 Set up Grok Bot + skills — about 30 min

Huddle **about 5 min**

Everyone opens aiinaction.up.railway.app/grokbot **about 10 min**

The driver pastes the setup prompt below **about 10 min**

Smoke test passes **about 5 min**

DONE LOOKS LIKE

Every member signed in; the driver's conversation has the skills loaded and has read one practice file; capture log started

FACILITATORS CIRCULATE

Check every laptop is signed in and the smoke test shows a file marked SYNTHETIC. GitHub blocked? Point to the starter bundle on /grokbot.

2 Warm-up missions — about 40 min

Split into two pairs **about 5 min**

Run both missions in parallel **about 20 min**

Push on it **about 10 min**

Swap outputs **about 5 min**

DONE LOOKS LIKE

Field insights (safety finding raised first) and an enquiry register exist; the team agreed one shared gap; 3 observations logged

FACILITATORS CIRCULATE

Let the missions finish without steering. Watch for the safety stop and the human-review moment; keep the 2-minute swap on time.

3 Workflow ideation — about 45 min

Map today's workflow for one real gap-to-content cycle **about 15 min**

Spot the AI ideas **about 10 min**

Pick one problem **about 10 min**

Sketch the swarm on paper **about 10 min**

DONE LOOKS LIKE

The gap-to-content cycle mapped; one problem chosen; hours per cycle and number of people involved estimated; ideas marked Efficiency or Opportunity

FACILITATORS CIRCULATE

Ask the ideation questions (page 6). Push for one problem statement and a baseline estimate before they sketch the swarm.

4 The hack — about 80 min

Write the assignment **about 10 min**

Run 1 **about 35 min**

Optional stretch break **about 5 min**

Run 2 **about 20 min**

Capture **about 10 min**

DONE LOOKS LIKE

Gap table with source IDs, persona × stage matrix and draft content plan with MLR pack, stopping at both gates; safety list first; screenshots and timings captured

FACILITATORS CIRCULATE

Ask the hack, gate and data questions. Near the end of the afternoon, check that screenshots and the capture log are saved.

DAY 2 · WED OCT 14 · FINAL BUILD → PRESENTATIONS IN ORDER 01→04

Final build + rehearsal **about 30–40 min**
re-open → last fix, agent fills the template → team review → rehearse → MC opens

01 Publications Brain **about 20 min**

02 Insights Engine **about 20 min**

03 Congress Monitor **about 20 min**

04 Field Intelligence Engine **about 20 min**

Then WPP demo

Insights Engine presents second of four, about 20 minutes, give or take, including a short live demo and a few questions. Keep all decks on one laptop and treat the capture step near the end of Day 1 as essential.

Warm up, then pick your data

Warm-up missions

Block 2 · about 40 min

Level 2 · Pair — **Pair A**

"What should leadership know?"

Field observations to insights. Watch for the safety finding the agent should raise before any analysis, then ask which insights point to a care or knowledge gap.

aiinaction.up.railway.app/missions/**field-insights**

Level 1 · Starter — **Pair B**

"Handle the enquiry queue"

The enquiry queue is a second gap signal: what HCPs keep asking tells you what they do not yet know.

aiinaction.up.railway.app/missions/**medical-information**

STRETCH, ALTERNATIVES AND REFERENCES

L1 Starter · L2 Pair · L3 Team · L4 Swarm

L1 Alternative
Turn the call notes into useful follow-up
/missions/msl-post-call

L3 Hack reference
Work across a practice CRM and content library
/missions/connected-planning

L3 Optional stretch
Run the next 30 days of Medical Affairs
/missions/thirty-day-capstone

L4 Also useful
Build the whole medical launch plan
/missions/launch-plan-swarm

BG Background
Team mission 2: the original team mission
Medical-Affairs-Skills · workshop

Data pack

browse and copy links at aiinaction.up.railway.app/data

Fictional practice data, or bring your own non-confidential data.

Nordvant Biopharma and its products are fictional. Own data: no patient data, no confidential company data, no real HCP personal data.

PICK ONE TA — MM · NORVANTIB · AD · DERMALYX · Obesity · ADIPOSYN

Any TA; synthetic data (fictional Nordvant Biopharma) with seeded safety findings and a promotional objective to catch.

TA PACK FILES synthetic/<TA>/

- TA field-observations.csv
- TA medical-plan.md
- TA medical-education-needs.md
- TA medical-information-enquiries.csv
- TA interaction-notes.md
- TA guideline-landscape.md
- TA evidence-landscape.md
- TA mlr-review-comments.md

PRACTICE CRM synthetic/connected/

- CRM hcps.csv
- CRM interactions.csv
- CRM engagement.csv
- CRM content_assets.csv
- CRM enquiries.csv
- CRM data-dictionary.json

Synthetic (fictional Nordvant Biopharma). hcps.csv gives setting and channel for personas; content_assets.csv is the library (audience, format, review status). Keep source IDs (e.g. OBS-001) on every gap.

Efficiency vs Opportunity, and the digital team

Efficiency AI today's work, faster

E1 Field observations, enquiries and the needs assessment coded into one gap list with source IDs, in minutes instead of weeks.

E2 HCPs segmented by persona and knowledge stage automatically, with the reasons for each assignment visible.

E3 A first-draft content plan that reuses what is already in the library before anything new is written, ready for MLR with the claims and references listed.

vs

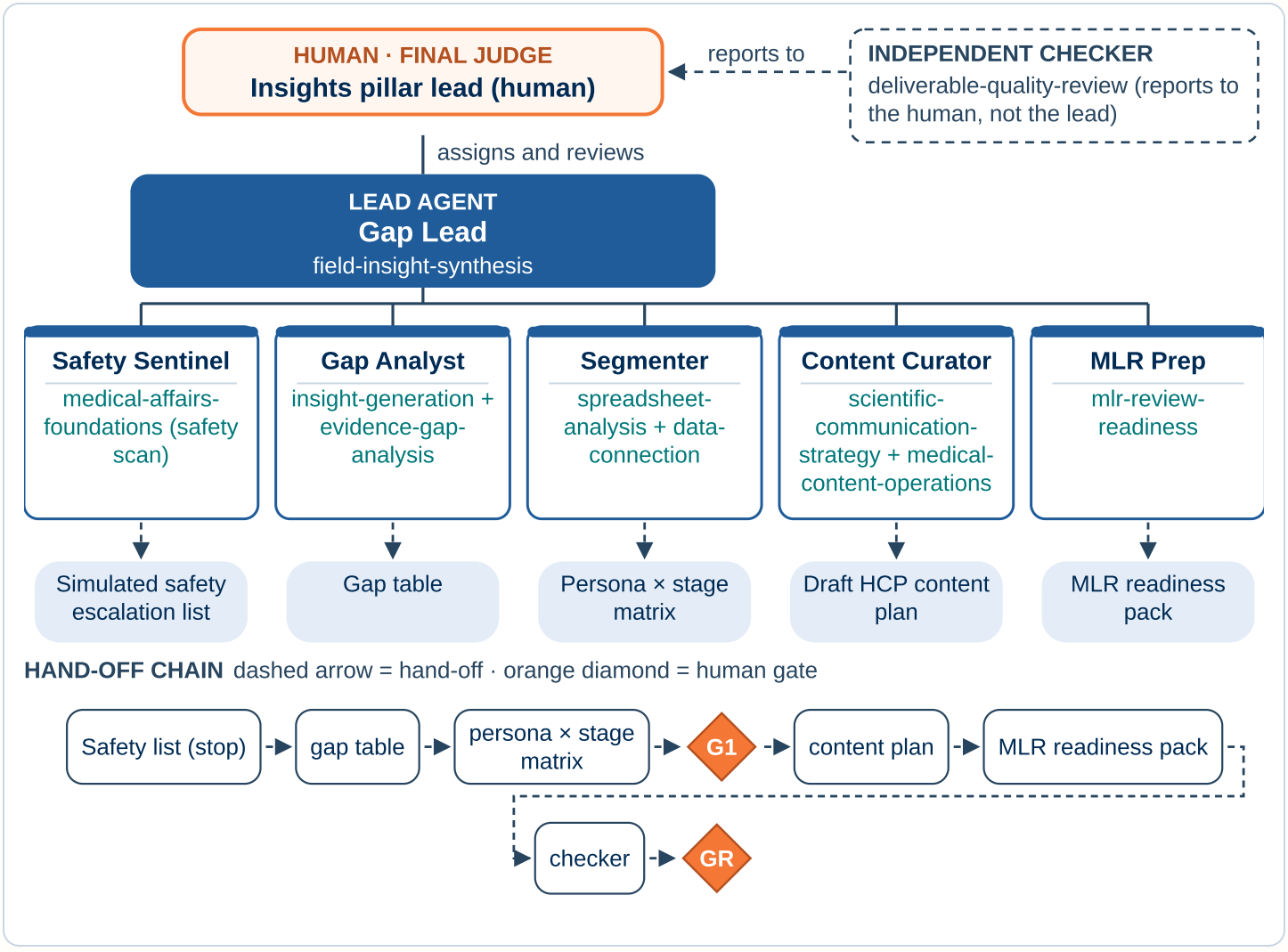
Opportunity AI something not possible today

01 Content that closes a named gap: every item in the plan traces back to the insights and records that justified it.

02 The right depth for the right HCP: a community specialist meeting the data for the first time gets a different piece than an academic expert.

03 A closed loop: next quarter's insights show whether the gap narrowed, so the content plan is measured on knowledge and care, not on volume.

Suggested swarm one lead agent, sub-agents with their skills, two human gates



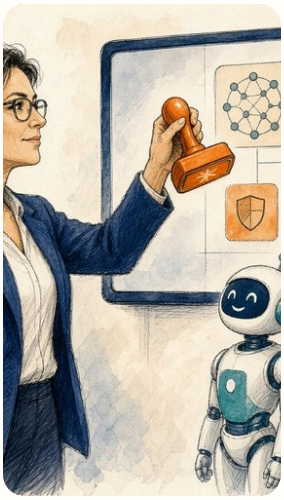
Where humans decide, and what to ask

G1 Gate 1

WHEN	After the Segmenter
WHO DECIDES	**The insights pillar leads**
WHAT THEY CHECK	Approve the gaps, the persona × stage segments and the safety routing before content is curated.
WHY A HUMAN	What counts as a gap is a judgment; safety routing is regulated.

G2 Gate 2

WHEN	After MLR Prep
WHO DECIDES	**MLR reviewers**
WHAT THEY CHECK	Approve each new or adapted item and the medical lead signs off the plan before anything reaches an HCP.
WHY A HUMAN	Accountability for what HCPs are told; MLR is regulated.



Always-on stop (a rule, not a gate): any possible adverse event, product quality complaint or off-label signal in human-sourced records is listed verbatim and routed by a human before analysis continues.

Facilitator question bank

the Day 1 feedback loop: ask one or two per visit, then move on

Ideation block 3

1. Where do care and knowledge gaps come from in your team today: field insights, medical information, needs assessments? Who joins them up?
2. How do you tell a knowledge gap (HCPs do not know) from a care gap (they know but practice differs)? Does the answer change the content?
3. Which HCP personas matter most for this TA, and how would you know what knowledge stage each one is at?
4. How much of your current content is reused, and how do you know which gap a piece was meant to close?

The hack block 4

5. Pick one persona × stage cell. Why that content, format and channel? Was existing content reused first?

Gates during the hack

6. Did the swarm catch the promotional learning objective in the needs assessment? Where does the medical vs promotional line sit in your plan?

Data during the hack

7. Did your swarm raise the seeded safety finding before anything else? Who receives it, and what does the agent stop doing?
8. Show me one gap and the record IDs behind it. Is it a care gap or a knowledge gap, and how confident is the swarm?
9. How did the swarm assign a knowledge stage to an HCP? What evidence did it use, and what would change the assignment?

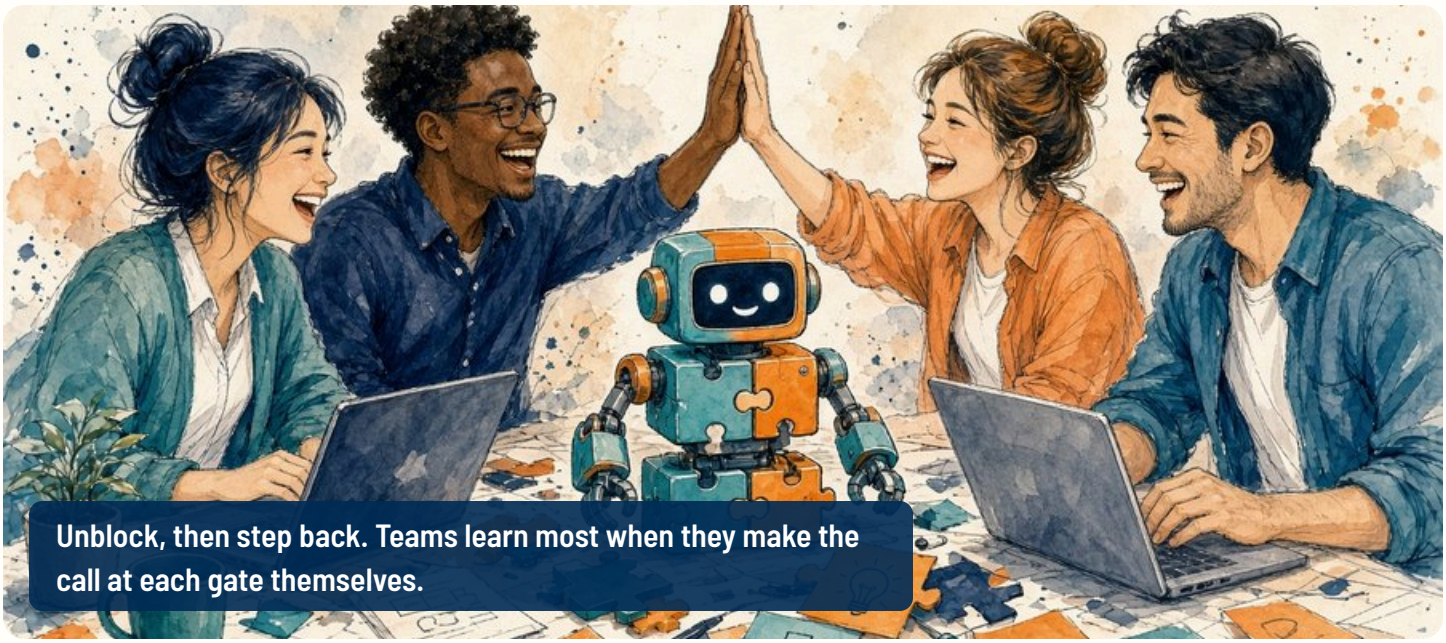
Impact during the hack

10. How many hours does one gap-to-content cycle take today, across everyone involved? What does the AI-assisted cycle take, including review and MLR?

When a team gets stuck

The ten snags facilitators see most often in this vertical, each with the unblocker to offer. Nudge, then step back: the team does the building.

Pitfall	How to unblock
01 Insights, not gaps.	Ask: "What do HCPs not know or not do, which HCPs, and how do we know?" Load evidence-gap-analysis.
02 The safety scan is skipped or buried.	Make the Safety Sentinel the first worker and tell the lead to stop until a human acknowledges the list.
03 Personas made up rather than drawn from the data.	Build personas from hcps.csv (setting, channel); show the count in every cell.
04 Everything becomes new content.	Curator checks content_assets.csv first: reuse, adapt, then create.
05 Promotional drift.	Keep objectives about competence and practice, not product use; MLR Prep flags claims and builds a readiness pack per item.
06 Too much data to read in the time.	Start with one TA and two personas; add the CRM tables and the full matrix in the second run.
07 Grok Bot cannot open GitHub.	Upload the starter bundle for your TA from https://aiinaction.up.railway.app/grokbot, or the repository ZIP.
08 The agent asks to connect company systems.	Say no: the workshop uses practice data. It is in every setup prompt.
09 A run is slow or wanders.	Ask for the plan first, then run one sub-agent at a time; cut the scope to one TA, one HCP, one output.
10 The output is generic.	Teach the agent: add 3–5 house rules (what good looks like, what must never appear) and run again.



Unblock, then step back. Teams learn most when they make the call at each gate themselves.

Impact worksheet and setup prompt

Impact worksheet the team's own estimates

One cycle = one gap-to-content planning cycle

Today baseline	h	×		×	$ /h	=	$
	hours per cycle (all people)		cycles per year		loaded rate		cost per year
			vs				
AI-assisted with the swarm	h	×		×	$ /h	=	$
	hours per cycle, incl. human review		cycles per year		loaded rate		cost per year

HOURS SAVED PER YEAR

(today h − AI h) × cycles = h

VALUE PER YEAR (ESTIMATE)

hours saved × rate = $

OTHER BENEFITS

days	%	#
speed to decision	coverage	errors caught

ILLUSTRATIVE **Not data, only an example:** 60 h → 12 h per cycle × 4 cycles/year = 192 h saved; × $150/h = $28,800 per year. Replace every number with the team's own estimate.

Count every person involved and include human review time in the AI-assisted row. These numbers go on the Impact slide of the final template (slide 9: formula tiles and a bar chart built from the team's own estimates).

Setup prompt Block 1, after sign-in · the driver pastes it into Grok Bot (replace the TA)

1 · Setup prompt copy all

```
Use https://github.com/Open-Medical-Affairs/Medical-Affairs-Skills
Read AGENTS.md. We are the Insights Engine team at AI in Action for Medical Affairs.
Load medical-affairs-foundations and these skills: field-insight-synthesis, insight-generation, evidence-gap-analysis, scientific-communication-strategy, medical-content-operations, mlr-review-readiness.
Use the practice data in https://github.com/Open-Medical-Affairs/Data-Sources: synthetic/[oncology-mm | immunology-ad | cardiometabolic-obesity]/ and synthetic/connected/ (the practice CRM). It is fictional (Nordvant Biopharma).
First, only: list the skills you loaded, then show the first 5 rows of field-observations.csv and confirm the file is labelled SYNTHETIC. Do not start a mission yet.
Do not ask me to connect company systems. Do not send, post or change anything outside this conversation.
```

SUGGESTED ROLES (ROTATE IF YOU LIKE)

Driver	Navigator	Scribe	Timekeeper + presenter
types into Grok Bot	reads the mission and pushes back	owns the capture log	keeps the clock, tells the story

02 COPYABLE PROMPT

Assignment starter for the hack

Paste into Grok Bot at the start of Block 4 (the hack), then edit the [brackets]. The setup prompt for Block 1 is on page 8. Ready prompts for every mission: **aiinaction.up.railway.app/prompts** · swarm mode and the Skill creator: **/optimizer**.

2 · Assignment starter: Insights Engine swarm — **Block 4 · start of the hack**

```
# Assignment: Insights Engine swarm
You are the Gap Lead, a lead agent coordinating a small team of digital workers for a Medical Affairs team.
This is an assignment, not a question: plan, delegate, check and hand back finished drafts.

## Goal: what done looks like
[One sentence: the problem we picked in ideation, and the deliverable that solves it.]

## Context to use
- https://github.com/Open-Medical-Affairs/Medical-Affairs-Skills : read AGENTS.md, load medical-affairs-
foundations, your lead skill field-insight-synthesis, and the skills named below.
- Practice data (fictional): https://github.com/Open-Medical-Affairs/Data-Sources synthetic/[TA]/ and
synthetic/connected/. Or our own non-confidential data: [describe].

## How to work
1. Show me a short plan and the org chart of workers first. Wait for my OK.
2. Scan human-sourced records for possible safety findings (adverse events, product complaints, off-label
use) before anything else. List them verbatim and stop until I acknowledge.
3. Run these sub-agents, each with only the context it needs:
   - Safety Sentinel (skills: medical-affairs-foundations (safety scan)): Reads every record first; lists
possible AEs, PQCs and off-label mentions verbatim and stops for routing. Hands off: Simulated safety
escalation list.
   - Gap Analyst (skills: insight-generation + evidence-gap-analysis): Turns field insights, enquiries and
the needs assessment into care and knowledge gaps with source IDs and confidence. Hands off: Gap table.
   - Segmenter (skills: spreadsheet-analysis + data-connection): Groups HCPs by persona and knowledge stage
and maps each gap to the cells it affects. Hands off: Persona × stage matrix.
   - Content Curator (skills: scientific-communication-strategy + medical-content-operations): Matches each
cell to the content library: reuse, adapt or create, with format and channel. Hands off: Draft HCP content
plan.
   - MLR Prep (skills: mlr-review-readiness): Lists claims, references and fair-balance points for every new
or adapted item; flags anything promotional. Hands off: MLR readiness pack.
4. STOP at Gate 1: After the Segmenter: the insights pillar leads approve the gaps, the persona × stage
segments and the safety routing before content is curated. Show me the evidence pack and wait.
5. Continue, then have an independent checker (deliverable-quality-review (reports to the human, not the
lead)) review the draft. Fix what it finds.
6. STOP at Gate 2: After MLR Prep: MLR reviewers approve each new or adapted item and the medical lead signs
off the plan before anything reaches an HCP. Wait for my decision.
7. Keep a timing note: minutes per step, and minutes of human review at each gate.

## House rules
- Never invent a citation, number, quote or meeting fact. Unsupported claims are marked and left for a human.
- Keep source IDs on every claim. Keep public evidence separate from fictional practice data.
- Do not send, post, publish or change any external system. Drafts only. Mark every output DRAFT and the data
FICTIONAL.
```

All products, people and results in the practice data are fictional. Outputs are drafts that need qualified human review; this is not a clinical decision, GxP or pharmacovigilance system.